Supporting Information for When are incomplete river-temperature records fit for estimating thermal extremes?

AUTHOR ONE[1]

[1]AFFILIATION PENDING AUTHOR CONFIRMATION

Contents

Text S1–S9; Figures S1–S2; generated Tables S1–S24. Full case-level and split-level results accompany the table-source CSV files; summary tables do not replace their denominators.

S1 Reference custody and exclusions

The opened-source reference and all old donor columns were inventoried before defining new eligibility. Identity normalization uses provider and original identifier; candidate aliases use coordinates and available polygon connections. Shared records connect network labels into overlap components. Old catchment polygons are incomplete, so European station disjointness is not universal hydrological disjointness. USGS-only transfer provides an additional river-flow-disjoint source restriction.

The HydroShare resource contains extremes, meteorology, flow and catchment attributes, not continuous daily water-temperature truth. Continuous Swiss and French values were acquired separately, after the salted role assignment. Metadata and completeness can define eligibility but recovery error cannot. Ineligible components are excluded without redistributing their roles. Swiss source release states are filtered in the acquisition contract. French date and time fields are combined before local-day coverage is checked; qualification 4 is explicitly unqualified, not validated. Thus French values are project aggregates and are not promoted into the primary documented-mean scope.

The audited Austrian route was monthly-only, and the Norwegian continuous route remained unresolved; neither supplies continuous truth. Unused USGS catalogs contain metadata only and were not converted into further acquisition

after source outcomes. The final new sample has 53 stations, with 21, 13 and 19 development/calibration/test stations. The prospective evaluation actually requires complete 2019–2022 target years and is smaller, especially in France. Raw counts appear in every result rather than assigning all nineteen test stations to every assay.

S2 Distinct experimental populations

Controlled recovery crosses feasible climatological overlap and length within the same reference year. Four realizations average placement noise; they do not multiply the number of independent histories. Warm controls restrict total missing warm-season days, while fragmentation holds their total budget but need not hold true support exposure fixed. Missing cells and fewer-than-three permitted training-year exclusions remain in job manifests.

The admission cohort uses short as well as long gaps and up to five empirical geometries, not the balanced long-gap mechanism cohort. The change was written before new calibration/test outcomes. The earlier controlled-classifier results remain archived diagnostic evidence; their larger gains are not used as primary admission findings.

Recorded-calendar replay transfers actual incomplete-year geometries to complete years. It is conditional on availability of qualifying incomplete calendars and complete counterpart years, not a census of all monitoring years. New qualifying replay calendars are French aggregates; no Swiss documented-mean natural-calendar estimate is available. Neither missing-day mechanisms nor temperatures originally hidden by the monitoring system are identified.

S3 Models, information and uncertainty

Matched quantile boosting uses 350 iterations, depth four, learning rate 0.06, minimum leaf size forty and quantile 0.9. Generic features contain gap geometry, season, visible donors, training climatological variability, historical record length, boundary observations, air availability and H30-estimator disagreement. Actual reconstruction OOF residual diagnostics, mask-matched historical H30 errors and direct H30 uncertainty are stronger decision-time alternatives, not hidden evaluation truth. Full window features add probabilistic support, its spread, peak-start

spread/entropy, competition and interactions. Exact allowlists are in the analysis contracts; truth, true overlap and realized errors are forbidden predictors.

Spline models add a five-knot cubic basis to the same small information set. The two-point equivalence margin is distinct from five-point superiority. Conditional source intervals lying inside the former do not demonstrate whole-procedure or cross-population equivalence.

Residual trajectories preserve observed target values. Training residuals are generated by holding out each permitted historical year as well as the current target year. Fourteen-day blocks cannot cross NaN boundaries. Short interpolation gaps may use complete blocks equal to their longest run only when no nominal block exists. Unsupported uncertainty is explicit missing information; it is not zero uncertainty. Ninety-percent scenario bands are not promised coverage and are evaluated as such.

Certificate references retain their declared minimum training support. Some locally fitted certificate heads are unavailable with the new development budget. Those are not successful zero-risk predictions and cannot establish an architectural win for another method. A selected certificate's delivered point and error differ from a certificate-derived ranking applied to the fixed priority point.

S4 Risk estimand and proof

For a fixed predictor and candidate threshold, let each deployment component contribute within-component mean admission A_g and failure-and-admission B_g. Since $0 \le B_g \le A_g \le 1$,

$$X_g = B_g + r(1 - A_g) \in [0, 1], \qquad r = 0.05.$$

The desired ratio is $E[B_g]/E[A_g]$, not mean component-conditional risk and not arbitrary case-weighted risk. For G independent deployment-representative components, invert the bounded-variable Chernoff–KL inequality to obtain an upper mean bound U satisfying

$$G\,\mathrm{kl}(\bar{X}, U) = \log(m/\alpha),$$

where m is the number of training-fixed candidate thresholds. A union bound covers candidate search, including nonmonotone risk. If any threshold has $U \le r$ and

positive admission, then $E[B_g] \leq rE[A_g]$ on the simultaneous-validity event. No result follows under arbitrary target shift or dependent components. The zero-mean upper bound is $1 - (\alpha/m)^{1/G}$, strictly positive.

Even the ideal all-admit, zero-failure observation therefore needs 59 groups for one candidate or 118 for twenty-one. Target calibration is too small. Empty certified admission is insufficient evidence, not observed risk zero or a recommendation to discard every record.

The empirical bootstrap rule is a separate descriptive operating policy. Constructed populations include twelve, forty and one hundred perfectly dependent-within-group cases and a rare high-risk group, with uninformative scores. These are counterexamples to general calibration claims, not measured frequencies in rivers. Test intervals describe variation over components conditional on fixed predictors; they cannot supply probability for an unobserved failure type.

S5 Freezing and time

Comparative source folds are 60/20/20 fitting/calibration/testing. Final source heads use an independent source calibration split, fitting on the other source components. F0 transfers the head and source threshold; F1 changes only target calibration; F2 trains on new development records and calibrates on the target calibration role. Provider-excluded and USGS-only restrictions remain distinct. Historical local reconstruction fitting is allowed by the stated deployment scenario.

The method lock predates target calibration errors, and the deployment lock predates test errors. Every predictive source file, fitted object and raw new daily input is hashed. No score, threshold, outcome margin or salted role is changed after testing. The fully prospective global head/calibration use only years through 2018; test years are 2019–2022 and every upstream local input uses earlier years. Earlier-year local reconstruction with all-calendar source cross-fitting is labelled Past-local, not fully prospective.

Prospective in compact table labels denotes calendar-time forward hindcasting. Inputs are current-vintage finalized observations and reanalysis; their original publication/revision vintages were not archived. The experiment does not establish actual as-of-2018 data availability or real-time field deployment. Predictor and local

training dates remain strictly earlier; no frozen prediction changes to make this qualification.

S6 Reconstruction references

The CSDI reference has two layers, two heads, 32 channels, fifty diffusion steps, a 366-step single feature, mixed random/structured masks, maximum thirty epochs and patience five. It is trained on documented old-source records only, with independent old validation groups and one metadata-chosen validation station-year per group. Target normalization excludes the evaluated year. Eight conditional draws provide a pointwise median series and a secondary median of trajectory H30 values. It is a fixed retrospective reference, not optimized state of the art or a prospective model.

Air2stream-3/8 use the existing published-equation implementation with compiled numerical kernels verified against it. Parameters fit the longest permitted contiguous forcing segment excluding the target year, and strictly before it prospectively. Discharge normalization uses training forcing only. Each interior gap starts from the last observed left anchor, never hidden truth. Incomplete forcing, unavailable discharge, insufficient training and nonfinite predictions are explicit statuses. Optimizer status and training error are retained. Comparisons show input requirements, availability and fixed-policy error on exactly the same supported cases.

S7 Decisions, unknowns and representation

All operating policies begin with an identical candidate pool within an assay. Unknown or unsupported answers remain in its denominator. Local hot thresholds use permitted training-year H30 quantiles; fixed 18 and 20-degree sensitivity thresholds are not regulatory claims. Full false-positive/negative, correct/unknown hot/cool counts, near-threshold outcomes and network coverage are in original-pool CSVs.

The cost grid charges every unknown record and assumes no perfect repair. Intervals are conditional descriptive component-bootstrap intervals, not simultaneous inference across all costs or measured ecological values. Selected hot-frequency

error is distinct from original-pool unknown-label bounds. Those bounds allow all unknown labels to be cool or hot and do not correct mistakes in reported labels.

A post-test deterministic baseline audit adds always-unknown, always-hot and always-cool policies to the unchanged cost grid. Always unknown has loss exactly the abstention cost; it produces no magnitude prediction and no finite selected-risk estimate. This does not alter the original H3 comparator, but prevents an improvement over a weak reconstruct-all reference being presented as operational superiority. Point dominance in selected sensitivity cells is not a new confirmatory endpoint.

The largest fully observed window mean is a deterministic lower bound on the complete recorded maximum. It can retain a hot label without accepting H30 magnitude, but cannot establish a cool label. An eighty-percent mean is not a bound. Fixed observation margins are sensitivity scenarios, not provider error specifications. The extension changes no magnitude error event.

S8 Power, stopping and a nonpredictive correction

Each protocol has 1,000 complete learner/calibration refits. Entire overlap components are resampled, preserving all station histories and masks. Deterministic local fits are cached because their inputs do not change under whole-component resampling; global heads and thresholds are refitted. Evaluation components are also resampled for whole-procedure intervals. Learning curves use twenty subsets per non-full size; full-pool seed repeats have one distinct subset.

Power projections compare predeclared five-, eight- and ten-point alternatives with zero and five-point null margins. They assume source-like group distributions and do not establish common-risk feasibility. An effect exactly at a superiority boundary has only size-level rejection, not high power. Training/calibration uncertainty limits projections even with more test groups. No additional stations were acquired until a preferred method won.

An initial descriptive label mistakenly treated snow percentages as fractions. Correcting majority snow to fifty percent, with glacier influence first and unknown attributes explicit, affects no feature, fitted model, calendar, eligibility, component or frozen role. Original locks and outputs remain unchanged. Script 373 supplies corrected descriptive subgroup tables; raw script-360 hydrological labels are super-

seded. A descriptive near-natural string in an earlier case manifest is not verified regulation information and supports no unregulated-catchment claim. The dated erratum records when this correction was made.

S9 Generated evidence tables

Tables are generated by script 369 from result rows. Full case/seed outputs, unsupported head statuses, exclusions, fingerprints and table-source CSVs are authoritative when a compact table suppresses detail. Dashes mean unavailable or nonestimable, never demonstrated zero risk.

Table S1: Opened-source admission cohort. Network labels are not independent when target or donor records overlap. Documented daily means define the primary measurement scope; unclear daily statistics and project-aggregated records are sensitivity analyses.

Provider	Daily statistic	Labels	Stations	Cases
arso	undocumented	12	47	4394
chmi	project_aggregated_from_subdaily	17	73	8849
foen	daily_mean	2	11	3131
gkd_bayern	daily_mean	9	53	13818
hubeau	project_aggregated_from_subdaily	4	9	801
lubw	daily_mean	7	20	2223
nve_hydapi	daily_mean	17	41	3013
rws_waterwebservices	project_aggregated_from_subdaily	2	5	359
syke	non_daily_reference_reading	2	3	771
usgs	daily_mean	121	319	32652

Table S2: New-to-project continuous reference records after all-old-target/donor and metadata-only catchment exclusions. Roles were salted before daily values were acquired and never reshuffled. French daily means are project aggregates and include explicitly unqualified source observations. Institutional novelty and station novelty are not the same.

Provider	Role	Stations	Components
foen	calibration	9	9
foen	development	11	11
foen	test	7	7
hubeau	calibration	4	4
hubeau	development	10	10
hubeau	test	12	12

Table S3: Primary documented-mean controlled cells. Four start realizations are attempted per feasible station-year cell; an impossible cell is not fabricated. Case counts do not measure independent replication.

Climate overlap	Days	Masks	Station-years
full	30	19536	4720
none	30	19536	4720
partial	30	19536	4720
full	60	19536	4720
none	60	19536	4720
partial	60	19536	4720
full	90	19536	4720
none	90	19536	4720
partial	90	19536	4720

Table S4: Paired full-minus-zero climatological overlap changes in absolute H30 error, averaging start realizations within a station-year before component bootstrap. Warm-season controls hold total missing warm-season days fixed. Intervals are 95 percent component-bootstrap intervals, not causal effects of an unobserved missingness mechanism.

Method	Control	Days	Change C	Lower	Upper	Groups
Daily climate	All seasons	30	0.622	0.584	0.656	144
Daily climate	All seasons	60	0.836	0.775	0.894	144
Daily climate	All seasons	90	0.896	0.824	0.964	144
Smooth climate	All seasons	30	0.628	0.589	0.669	144
Smooth climate	All seasons	60	0.860	0.800	0.920	144
Smooth climate	All seasons	90	0.926	0.849	0.996	144
Interpolation	All seasons	30	0.411	0.385	0.443	144
Interpolation	All seasons	60	1.032	0.980	1.090	144
Interpolation	All seasons	90	1.785	1.700	1.867	144
Neighbour ridge	All seasons	30	0.280	0.240	0.327	127
Neighbour ridge	All seasons	60	0.377	0.332	0.430	126
Neighbour ridge	All seasons	90	0.407	0.359	0.465	124
Air ridge	All seasons	30	0.516	0.449	0.593	113
Air ridge	All seasons	60	0.753	0.674	0.847	113
Air ridge	All seasons	90	0.817	0.733	0.914	113
Daily climate	Warm season	30	0.564	0.531	0.599	143
Daily climate	Warm season	60	0.678	0.637	0.719	143
Daily climate	Warm season	90	0.694	0.559	0.811	28
Smooth climate	Warm season	30	0.571	0.533	0.612	143
Smooth climate	Warm season	60	0.703	0.658	0.756	143

Method	Control	Days	Change C	Lower	Upper	Groups
Smooth climate	Warm season	90	0.695	0.558	0.807	28
Interpolation	Warm season	30	0.371	0.342	0.407	143
Interpolation	Warm season	60	0.904	0.847	0.971	143
Interpolation	Warm season	90	1.464	1.055	1.858	28
Neighbour ridge	Warm season	30	0.256	0.216	0.304	126
Neighbour ridge	Warm season	60	0.319	0.276	0.370	124
Neighbour ridge	Warm season	90	0.469	0.327	0.621	24
Air ridge	Warm season	30	0.458	0.392	0.538	113
Air ridge	Warm season	60	0.614	0.529	0.721	113
Air ridge	Warm season	90	0.796	0.587	0.995	20

Table S5: Paired 90-minus-30-day H30 error at high and fully hidden true support. The 0.999 cutoff denotes the exact all-hidden discrete case; the comparison remains conditional on feasible masks and eligible station-years. Length is not negligible for every reconstruction.

Method	Hidden cutoff	Change C	Lower	Upper	Station-years	Groups
Daily climate	0.900	0.147	0.121	0.172	2454	144
Daily climate	0.999	0.133	0.097	0.166	396	98
Smooth climate	0.900	0.159	0.132	0.185	2454	144

Method	Hidden cutoff	Change C	Lower	Upper	Station-years	Groups
Smooth climate	0.999	0.145	0.106	0.179	396	98
Interpolation	0.900	1.304	1.225	1.391	2454	144
Interpolation	0.999	1.235	1.120	1.347	396	98
Neighbour ridge	0.900	0.043	0.035	0.053	1685	120
Neighbour ridge	0.999	0.032	0.016	0.051	251	76
Air ridge	0.900	0.138	0.114	0.169	1889	113
Air ridge	0.999	0.123	0.093	0.156	313	76

Table S6: Fragmented-minus-contiguous contrasts with equal total missing warm-season days. True hidden support is not held fixed; its change is shown rather than silently attributing the entire effect to fragmentation. Policy eligibility uses longest run, not total missing days.

Method	Days	Error change C	Support change	Lower	Upper
Daily climate	30	-0.271	-0.254	-0.288	-0.252
Daily climate	60	-0.262	-0.273	-0.286	-0.238
Daily climate	90	-0.246	-0.260	-0.274	-0.218
Smooth climate	30	-0.274	-0.254	-0.293	-0.254
Smooth climate	60	-0.267	-0.273	-0.291	-0.240
Smooth climate	90	-0.253	-0.260	-0.279	-0.225
Interpolation	30	-0.242	-0.254	-0.257	-0.229
Interpolation	60	-0.597	-0.273	-0.638	-0.556
Interpolation	90	-1.395	-0.260	-1.485	-1.304
Neighbour ridge	30	-0.115	-0.253	-0.135	-0.097
Neighbour ridge	60	-0.105	-0.272	-0.122	-0.090
Neighbour ridge	90	-0.101	-0.251	-0.118	-0.088
Air ridge	30	-0.213	-0.248	-0.243	-0.187
Air ridge	60	-0.231	-0.271	-0.259	-0.205
Air ridge	90	-0.217	-0.256	-0.249	-0.191

Table S7: Exact decomposition and sign audit. A positive switch increment is often compensation for underestimation, not a fabricated hot peak. Numerical identity residuals below the displayed precision are not evidence of physical truth accuracy.

Method	Cases	Switches	Still low share	Max residual C
Daily climate	312825	188204	0.604	0.000
Smooth climate	312825	188373	0.632	0.000
Interpolation	312825	169266	0.710	0.000
Neighbour ridge	225103	86886	0.550	0.000
Air ridge	239231	125468	0.707	0.000

Table S8: Two-by-two residual intervention: original, permuted, mean removed, and both. The interaction is original minus permuted minus centered plus both. Truth is required for all these interventions. Their percentage improvements cannot be added as identified physical contributions. Deployable visible-anchor adjustment is evaluated separately in the text.

Method	Intervention	MAE / interaction C	Lower	Upper	Groups
Daily climate	raw	0.450	0.412	0.486	144
Daily climate	permuted	0.398	0.370	0.425	144
Daily climate	demeaned	0.215	0.193	0.236	144
Daily climate	demeaned_permuted	0.071	0.068	0.075	144
Daily climate	interaction	-0.091	-0.101	-0.079	144
Smooth climate	raw	0.459	0.420	0.496	144
Smooth climate	permuted	0.397	0.368	0.424	144
Smooth climate	demeaned	0.221	0.200	0.243	144
Smooth climate	demeaned_permuted	0.069	0.065	0.073	144

Method	Intervention	MAE / interaction C	Lower	Upper	Groups
Smooth climate	interaction	-0.091	-0.101	-0.079	144
Interpolation	raw	0.521	0.497	0.545	144
Interpolation	permuted	0.461	0.440	0.483	144
Interpolation	demeaned	0.223	0.211	0.235	144
Interpolation	demeaned_permuted	0.072	0.069	0.075	144
Interpolation	interaction	-0.092	-0.098	-0.085	144
Neighbour ridge	raw	0.186	0.164	0.215	128
Neighbour ridge	permuted	0.165	0.144	0.192	128
Neighbour ridge	demeaned	0.076	0.066	0.087	128
Neighbour ridge	demeaned_permuted	0.027	0.024	0.030	128
Neighbour ridge	interaction	-0.028	-0.034	-0.023	128
Air ridge	raw	0.388	0.350	0.433	113
Air ridge	permuted	0.318	0.292	0.345	113
Air ridge	demeaned	0.172	0.157	0.191	113
Air ridge	demeaned_permuted	0.047	0.043	0.051	113
Air ridge	interaction	-0.055	-0.062	-0.047	113

Table S9: Same-point source admission with disjoint 60/20/20 fitting, calibration and evaluation components. Entries are median [minimum, maximum] over three split seeds, not confidence intervals or independent experiments. Past-local recomputes local inputs from earlier years but does not freeze the global head in calendar time. All scores share the same empirical 5 percent calibration target, not identical realized test risk.

Score	Protocol	Retention percent	Selected failures percent
Certificate proxy	Retro	72.7 [72.6, 73.7]	2.6 [2.5, 2.9]
Complete diag.	Retro	78.8 [78.1, 79.0]	2.7 [2.6, 2.7]
Daily disagreement	Retro	1.0 [0.0, 1.8]	4.1 [3.3, 4.9]
Raw H30 uncertainty	Retro	77.8 [77.2, 79.3]	2.8 [2.6, 3.2]
Generic + OOF	Retro	78.1 [78.0, 79.1]	2.7 [2.6, 3.0]
Generic daily	Retro	71.2 [70.2, 72.4]	3.0 [2.8, 3.2]
Direct H30	Retro	78.9 [77.8, 79.6]	2.7 [2.5, 3.0]
Generic + overlap	Retro	76.4 [75.5, 77.7]	2.9 [2.6, 3.3]
Generic H30	Retro	76.6 [74.0, 77.4]	3.0 [2.4, 3.3]
Historical H30	Retro	72.8 [72.8, 73.6]	2.8 [2.8, 2.9]
Length	Retro	36.8 [36.7, 37.9]	1.9 [1.8, 1.9]
OOF daily scale	Retro	32.6 [24.3, 36.9]	2.4 [1.6, 3.0]
Climate overlap	Retro	67.6 [66.9, 68.7]	3.0 [2.9, 3.1]
Policy certificate	Retro	51.1 [50.7, 51.3]	2.4 [2.4, 2.6]
Mean support	Retro	69.7 [68.6, 69.9]	2.7 [2.4, 2.7]
Overlap x residual	Retro	75.5 [74.3, 75.7]	3.1 [2.7, 3.1]
Season spline	Retro	72.6 [71.9, 74.1]	2.9 [2.9, 3.3]
Spline + overlap	Retro	72.8 [71.6, 73.0]	2.9 [2.7, 3.0]
Small logit	Retro	64.2 [59.4, 66.6]	3.3 [2.8, 3.8]
Logit + overlap	Retro	69.9 [68.9, 69.9]	2.9 [2.9, 3.3]
H30 disagreement	Retro	70.8 [70.2, 71.5]	3.1 [2.8, 3.2]
Window	Retro	78.6 [77.8, 79.0]	2.6 [2.4, 2.8]
No competition	Retro	78.5 [77.7, 80.0]	2.6 [2.4, 3.1]
No donor features	Retro	79.5 [79.0, 79.8]	2.9 [2.8, 3.0]

Score	Protocol	Retention percent	Selected failures percent
No window spread	Retro	78.5 [77.8, 80.0]	2.6 [2.4, 3.1]
Upper support	Retro	69.9 [69.8, 70.0]	2.4 [2.4, 2.4]
Warm coverage	Retro	52.5 [49.7, 53.5]	2.2 [1.6, 2.4]
Certificate proxy	Past-local	69.4 [67.8, 69.7]	2.7 [2.5, 2.7]
Complete diag.	Past-local	76.0 [75.0, 77.5]	2.5 [2.4, 3.0]
Daily disagreement	Past-local	0.0 [0.0, 1.7]	7.8 [7.8, 7.8]
Raw H30 uncertainty	Past-local	74.4 [73.1, 74.8]	2.9 [2.6, 3.1]
Generic + OOF	Past-local	75.0 [74.0, 76.3]	2.5 [2.3, 2.9]
Generic daily	Past-local	66.0 [61.3, 66.2]	3.1 [2.5, 3.1]
Direct H30	Past-local	76.4 [75.7, 76.6]	2.7 [2.7, 2.9]
Generic + overlap	Past-local	72.0 [69.3, 73.5]	2.5 [1.9, 2.9]
Generic H30	Past-local	71.0 [69.0, 71.6]	2.2 [2.0, 2.6]
Historical H30	Past-local	62.9 [58.4, 64.1]	2.8 [2.1, 2.9]
Length	Past-local	37.4 [31.0, 37.5]	2.2 [1.8, 2.2]
OOF daily scale	Past-local	28.5 [21.7, 29.0]	2.3 [1.6, 3.0]
Climate overlap	Past-local	62.6 [59.5, 63.2]	3.0 [2.6, 3.4]
Policy certificate	Past-local	51.1 [48.4, 51.3]	2.4 [1.8, 2.5]
Mean support	Past-local	65.0 [62.6, 65.2]	2.7 [2.2, 2.9]
Overlap x residual	Past-local	69.9 [69.8, 70.1]	3.2 [3.0, 3.7]
Season spline	Past-local	70.1 [70.0, 70.7]	3.1 [3.0, 3.1]
Spline + overlap	Past-local	70.8 [70.1, 71.8]	3.2 [3.0, 3.5]
Small logit	Past-local	56.1 [55.9, 58.8]	3.0 [2.6, 3.5]
Logit + overlap	Past-local	62.8 [60.6, 63.6]	3.4 [2.7, 3.5]
H30 disagreement	Past-local	66.9 [65.9, 67.9]	2.7 [2.6, 2.9]
Window	Past-local	76.6 [75.1, 76.8]	2.7 [2.3, 2.7]
No competition	Past-local	76.6 [76.4, 76.7]	2.7 [2.7, 2.8]
No donor features	Past-local	77.2 [75.6, 77.7]	3.0 [2.4, 3.0]
No window spread	Past-local	76.4 [75.4, 76.6]	2.7 [2.4, 2.7]
Upper support	Past-local	65.5 [65.2, 67.4]	2.5 [2.4, 2.8]
Warm coverage	Past-local	48.1 [47.6, 48.3]	1.8 [1.7, 1.8]

Table S10: Conditional-on-fitted-procedure paired retention differences in percentage points. The primary superiority margin is 5 points and equivalence margin is 2 points. Crossing zero is not equivalence; a small conditional interval does not establish whole-procedure equivalence.

Comparison	Protocol	Seed	Difference	Lower	Upper
Window - Generic H30	Retro	101	1.698	0.247	3.034
Window - Direct H30	Retro	101	-0.501	-1.100	0.018
Window - Complete diag.	Retro	101	0.046	-0.669	0.722
Window - Generic + OOF	Retro	101	-0.026	-0.877	0.743
Generic H30 - Generic daily	Retro	101	6.176	5.100	7.361
Generic + overlap - Generic H30	Retro	101	0.350	-0.303	1.041
Logit + overlap - Small logit	Retro	101	9.541	6.882	12.776
Spline + overlap - Season spline	Retro	101	-1.247	-2.104	-0.486
Window - Generic H30	Retro	202	3.828	2.198	5.301
Window - Direct H30	Retro	202	-0.007	-0.271	0.261
Window - Complete diag.	Retro	202	-1.027	-1.850	-0.268
Window - Generic + OOF	Retro	202	-0.164	-0.771	0.492
Generic H30 - Generic daily	Retro	202	3.839	2.920	4.877
Generic + overlap - Generic H30	Retro	202	2.447	1.820	3.107
Logit + overlap - Small logit	Retro	202	5.662	3.275	8.309
Spline + overlap - Season spline	Retro	202	-1.012	-1.689	-0.394
Window - Generic H30	Retro	303	2.050	0.788	3.233

Comparison	Protocol	Seed	Difference	Lower	Upper
Window - Direct H30	Retro	303	-0.268	-0.676	0.139
Window - Complete diag.	Retro	303	0.527	-0.276	1.246
Window - Generic + OOF	Retro	303	0.467	-0.471	1.403
Generic H30 - Generic daily	Retro	303	4.132	2.723	5.716
Generic + overlap - Generic H30	Retro	303	-1.107	-1.658	-0.567
Logit + overlap - Small logit	Retro	303	3.366	1.543	5.522
Spline + overlap - Season spline	Retro	303	1.111	0.657	1.648
Window - Generic H30	Past-local	101	5.212	2.868	7.346
Window - Direct H30	Past-local	101	0.148	-0.435	0.885
Window - Complete diag.	Past-local	101	-0.768	-1.628	-0.093
Window - Generic + OOF	Past-local	101	0.472	-0.123	0.943
Generic H30 - Generic daily	Past-local	101	5.415	3.617	7.390
Generic + overlap - Generic H30	Past-local	101	1.900	1.041	2.827
Logit + overlap - Small logit	Past-local	101	4.689	3.245	6.226
Spline + overlap - Season spline	Past-local	101	1.725	0.611	3.131
Window - Generic H30	Past-local	202	7.622	5.526	9.306
Window - Direct H30	Past-local	202	0.241	-0.039	0.544
Window - Complete diag.	Past-local	202	1.566	0.513	2.601
Window - Generic + OOF	Past-local	202	2.624	1.297	4.385
Generic H30 - Generic daily	Past-local	202	7.721	6.608	9.039

Comparison	Protocol	Seed	Difference	Lower	Upper
Generic + overlap - Generic H30	Past-local	202	0.345	-0.116	0.775
Logit + overlap - Small logit	Past-local	202	7.477	6.148	9.026
Spline + overlap - Season spline	Past-local	202	0.825	0.270	1.451
Window - Generic H30	Past-local	303	4.149	1.464	6.589
Window - Direct H30	Past-local	303	-0.581	-1.442	0.338
Window - Complete diag.	Past-local	303	-0.861	-1.961	0.078
Window - Generic + OOF	Past-local	303	0.123	-0.628	0.839
Generic H30 - Generic daily	Past-local	303	4.971	3.224	6.699
Generic + overlap - Generic H30	Past-local	303	0.993	-0.502	2.481
Logit + overlap - Small logit	Past-local	303	4.009	2.538	5.589
Spline + overlap - Season spline	Past-local	303	-0.581	-1.213	-0.060

Table S11: Whole-procedure uncertainty from 1,000 network resamples, refitting four risk learners and recalibrating thresholds. The prospective global head and calibration use years through 2018 and evaluation uses 2019–2022. Cached local fits are unchanged under resampling their whole component.

Comparison	Protocol	Difference pp	Lower	Upper	Test groups
Window - Generic H30	Retro	2.595	-1.900	7.970	29
Direct H30 - Generic H30	Retro	2.535	-2.344	8.176	29
Window - Direct H30	Retro	0.059	-3.787	4.436	29
Direct H30 - Generic + OOF	Retro	1.800	-3.239	7.689	29

Comparison	Protocol	Difference pp	Lower	Upper	Test groups
Window - Generic H30	Prospective	2.916	-5.750	11.269	24
Direct H30 - Generic H30	Prospective	2.689	-6.952	10.955	24
Window - Direct H30	Prospective	0.226	-5.554	5.898	24
Direct H30 - Generic + OOF	Prospective	-1.466	-9.116	5.050	24

Table S12: Projected power to exceed the five-point practical margin under predeclared 8- and 10-point location-shift alternatives. Projections assume source-like independent groups and do not establish common-risk feasibility under transfer. Neither 80 nor 90 percent power is reached in this tested range; increasing only test groups leaves training/calibration variability. An effect exactly at the null margin cannot have high superiority power.

Protocol	Test groups	Assumed gain	Power
Retro	7	0.080	0.109
Retro	7	0.100	0.240
Retro	19	0.080	0.193
Retro	19	0.100	0.362
Retro	53	0.080	0.254
Retro	53	0.100	0.417
Retro	100	0.080	0.274
Retro	100	0.100	0.399
Retro	400	0.080	0.291
Retro	400	0.100	0.410
Prospective	7	0.080	0.084
Prospective	7	0.100	0.161
Prospective	19	0.080	0.086
Prospective	19	0.100	0.169
Prospective	53	0.080	0.127
Prospective	53	0.100	0.256
Prospective	100	0.080	0.122

Protocol	Test groups	Assumed gain	Power
Prospective	100	0.100	0.248
Prospective	400	0.080	0.136
Prospective	400	0.100	0.272

Table S13: Repeated source learning curves. Every non-full size uses twenty independently sampled subsets of the opened-source training pool; full-pool repeats vary learner seed but have one unique subset. Evaluation and calibration groups are fixed and independent of fitting. Standard deviations are across fits, not evidence from additional rivers.

Score	Protocol	Train groups	Subsets	Fits	Retention percent	SD pp
Generic + OOF	Retro	10	20	20	74.631	3.195
Direct H30	Retro	10	20	20	77.514	2.675
Generic H30	Retro	10	20	20	72.796	3.731
Window	Retro	10	20	20	77.495	2.388
Generic + OOF	Retro	25	20	20	77.619	1.557
Direct H30	Retro	25	20	20	78.981	1.881
Generic H30	Retro	25	20	20	76.417	1.675
Window	Retro	25	20	20	78.463	1.571
Generic + OOF	Retro	50	20	20	77.406	1.915
Direct H30	Retro	50	20	20	79.162	2.187
Generic H30	Retro	50	20	20	77.276	1.659
Window	Retro	50	20	20	79.334	1.981
Generic + OOF	Retro	75	20	20	77.860	0.750
Direct H30	Retro	75	20	20	79.541	0.840
Generic H30	Retro	75	20	20	79.009	1.195
Window	Retro	75	20	20	79.712	0.708
Generic + OOF	Retro	86	1	3	78.152	0.458
Direct H30	Retro	86	1	3	79.996	0.124
Generic H30	Retro	86	1	3	79.906	0.541

Score	Protocol	Train groups	Subsets	Fits	Retention percent	SD pp
Window	Retro	86	1	3	79.792	0.459
Generic + OOF	Prospective	10	20	20	72.627	2.837
Direct H30	Prospective	10	20	20	70.985	3.407
Generic H30	Prospective	10	20	20	69.027	5.188
Window	Prospective	10	20	20	71.321	3.455
Generic + OOF	Prospective	25	20	20	74.009	3.545
Direct H30	Prospective	25	20	20	74.054	1.954
Generic H30	Prospective	25	20	20	70.622	4.008
Window	Prospective	25	20	20	73.718	1.520
Generic + OOF	Prospective	50	20	20	74.667	1.786
Direct H30	Prospective	50	20	20	74.086	1.881
Generic H30	Prospective	50	20	20	71.285	2.883
Window	Prospective	50	20	20	73.835	2.349
Generic + OOF	Prospective	62	1	3	75.845	2.792
Direct H30	Prospective	62	1	3	74.882	2.476
Generic H30	Prospective	62	1	3	73.074	4.222
Window	Prospective	62	1	3	73.349	0.483

Table S14: Independent new-station transfer, with Swiss documented means primary and French project aggregates separate. F0: source model and source threshold unchanged; F1: same model, new local calibration; F2: new development fitting and local calibration; F0-P: target provider excluded; F0-US: USGS-only intercontinental source. A local reconstruction fit to permitted station history is allowed in every regime. The joint certificate delivers its own selected point; the other feature comparisons deliver the fixed information-priority point. None of these point risks is a per-network guarantee.

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Generic H30	Retro	foen	F0	51.121	5.848	7
Generic H30	Retro	foen	F1	45.740	3.268	7
Generic H30	Retro	foen	F2	43.946	3.061	7
Complete diag.	Retro	foen	F0	53.214	5.899	7
Complete diag.	Retro	foen	F1	45.291	2.640	7
Complete diag.	Retro	foen	F2	41.106	1.818	7
Direct H30	Retro	foen	F0	56.353	9.284	7
Direct H30	Retro	foen	F1	46.487	2.894	7
Direct H30	Retro	foen	F2	45.291	3.960	7
Window	Retro	foen	F0	56.054	8.800	7
Window	Retro	foen	F1	47.534	4.088	7
Window	Retro	foen	F2	44.395	3.030	7
Joint certificate	Retro	foen	F0	52.915	10.169	7
Joint certificate	Retro	foen	F1	47.085	6.984	7
Joint certificate	Retro	foen	F2	25.262	1.183	7
Generic H30	Retro	foen	F0-P	54.709	8.470	7
Direct H30	Retro	foen	F0-P	60.389	11.386	7
Window	Retro	foen	F0-P	59.791	11.250	7
Joint certificate	Retro	foen	F0-P	49.327	7.879	7
Complete diag.	Retro	foen	F0-P	55.306	6.757	7
Generic H30	Retro	foen	F0-US	52.167	6.877	7

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Direct H30	Retro	foen	F0-US	60.239	10.918	7
Window	Retro	foen	F0-US	56.951	9.711	7
Joint certificate	Retro	foen	F0-US	53.961	9.141	7
Complete diag.	Retro	foen	F0-US	55.605	6.989	7
Generic H30	Retro	hubeau	F0	48.857	1.462	12
Generic H30	Retro	hubeau	F1	59.143	3.623	12
Generic H30	Retro	hubeau	F2	55.000	3.377	12
Complete diag.	Retro	hubeau	F0	60.000	2.381	12
Complete diag.	Retro	hubeau	F1	52.714	1.355	12
Complete diag.	Retro	hubeau	F2	52.000	1.923	12
Direct H30	Retro	hubeau	F0	53.857	2.653	12
Direct H30	Retro	hubeau	F1	30.714	0.000	12
Direct H30	Retro	hubeau	F2	53.286	1.877	12
Window	Retro	hubeau	F0	69.143	6.198	12
Window	Retro	hubeau	F1	51.714	2.210	12
Window	Retro	hubeau	F2	46.286	2.160	12
Joint certificate	Retro	hubeau	F0	51.143	3.911	12
Joint certificate	Retro	hubeau	F1	18.429	0.000	12
Generic H30	Retro	hubeau	F0-P	47.571	1.502	12
Direct H30	Retro	hubeau	F0-P	54.286	2.895	12
Window	Retro	hubeau	F0-P	58.857	2.913	12
Joint certificate	Retro	hubeau	F0-P	55.000	5.195	12
Complete diag.	Retro	hubeau	F0-P	61.714	2.083	12
Generic H30	Retro	hubeau	F0-US	53.000	3.235	12
Direct H30	Retro	hubeau	F0-US	60.429	4.255	12
Window	Retro	hubeau	F0-US	61.429	4.651	12
Joint certificate	Retro	hubeau	F0-US	56.571	6.566	12
Complete diag.	Retro	hubeau	F0-US	63.571	2.921	12
Joint certificate	Retro	hubeau	F2	0.000	–	12
Generic H30	Prospective	foen	F0	43.353	6.667	7
Generic H30	Prospective	foen	F1	30.636	0.000	7

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Generic H30	Prospective	foen	F2	30.058	1.923	7
Complete diag.	Prospective	foen	F0	46.243	6.250	7
Complete diag.	Prospective	foen	F1	32.948	0.000	7
Complete diag.	Prospective	foen	F2	32.948	0.000	7
Direct H30	Prospective	foen	F0	54.913	14.737	7
Direct H30	Prospective	foen	F1	34.104	0.000	7
Direct H30	Prospective	foen	F2	15.029	0.000	7
Window	Prospective	foen	F0	47.399	9.756	7
Window	Prospective	foen	F1	36.994	1.562	7
Window	Prospective	foen	F2	13.873	0.000	7
Joint certificate	Prospective	foen	F0	53.179	9.783	7
Joint certificate	Prospective	foen	F1	28.902	2.000	7
Joint certificate	Prospective	foen	F2	4.046	0.000	7
Generic H30	Prospective	foen	F0-P	42.197	5.479	7
Direct H30	Prospective	foen	F0-P	38.728	2.985	7
Window	Prospective	foen	F0-P	43.931	7.895	7
Joint certificate	Prospective	foen	F0-P	38.150	1.515	7
Complete diag.	Prospective	foen	F0-P	49.133	8.235	7
Generic H30	Prospective	foen	F0-US	43.353	6.667	7
Direct H30	Prospective	foen	F0-US	41.618	6.944	7
Window	Prospective	foen	F0-US	41.618	6.944	7
Joint certificate	Prospective	foen	F0-US	31.214	5.556	7
Complete diag.	Prospective	foen	F0-US	36.994	3.125	7
Generic H30	Prospective	hubeau	F0	35.294	0.000	6
Generic H30	Prospective	hubeau	F1	42.353	2.778	6
Generic H30	Prospective	hubeau	F2	36.471	0.000	6
Complete diag.	Prospective	hubeau	F0	40.000	5.882	6
Complete diag.	Prospective	hubeau	F1	40.000	5.882	6
Complete diag.	Prospective	hubeau	F2	42.353	0.000	6
Direct H30	Prospective	hubeau	F0	44.706	10.526	6
Direct H30	Prospective	hubeau	F1	37.647	6.250	6

Score	Time	Provider	Mode	Keep percent	Fail percent	Groups
Direct H30	Prospective	hubeau	F2	44.706	5.263	6
Window	Prospective	hubeau	F0	49.412	7.143	6
Window	Prospective	hubeau	F1	42.353	5.556	6
Window	Prospective	hubeau	F2	38.824	0.000	6
Joint certificate	Prospective	hubeau	F0	36.471	16.129	6
Joint certificate	Prospective	hubeau	F1	0.000	–	6
Generic H30	Prospective	hubeau	F0-P	45.882	7.692	6
Direct H30	Prospective	hubeau	F0-P	34.118	0.000	6
Window	Prospective	hubeau	F0-P	42.353	5.556	6
Joint certificate	Prospective	hubeau	F0-P	40.000	5.882	6
Complete diag.	Prospective	hubeau	F0-P	49.412	4.762	6
Generic H30	Prospective	hubeau	F0-US	43.529	5.405	6
Direct H30	Prospective	hubeau	F0-US	40.000	11.765	6
Window	Prospective	hubeau	F0-US	43.529	5.405	6
Joint certificate	Prospective	hubeau	F0-US	42.353	13.889	6
Complete diag.	Prospective	hubeau	F0-US	34.118	0.000	6
Joint certificate	Prospective	hubeau	F2	0.000	–	6

Table S15: Corrected hydrological strata for the uncertain-window score. Snow precipitation is a percent: majority snow means at least 50 percent, not 0.5. Glacier influence has priority. This descriptive unit correction changed no predictor, threshold, case, eligibility or frozen role. Single-component strata cannot certify their risk; raw script-360 hydrological labels are superseded by this table.

Hydrology	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
glacier_influenced	Retro	foen	F0	68	16	1	–
other_mountain_catchment	Retro	foen	F0	159	9	3	0.086

Hydrology	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
snow_dominated	Retro	foen	F0	148	8	3	0.100
glacier_influenced	Retro	foen	F1	60	9	1	–
other_mountain_catchment	Retro	foen	F1	137	3	3	0.050
snow_dominated	Retro	foen	F1	121	1	3	0.026
glacier_influenced	Retro	hubeau	F0	84	0	2	0.000
other_mountain_catchment	Retro	hubeau	F0	400	30	10	0.132
glacier_influenced	Retro	hubeau	F1	71	0	2	0.000
other_mountain_catchment	Retro	hubeau	F1	291	8	10	0.056
glacier_influenced	Prospective	foen	F0	19	1	1	–
other_mountain_catchment	Prospective	foen	F0	30	1	3	0.091
snow_dominated	Prospective	foen	F0	33	6	3	0.364
glacier_influenced	Prospective	foen	F1	18	0	1	–
other_mountain_catchment	Prospective	foen	F1	22	0	3	0.000
snow_dominated	Prospective	foen	F1	24	1	3	0.091
other_mountain_catchment	Prospective	hubeau	F0	42	3	6	0.167
other_mountain_catchment	Prospective	hubeau	F1	36	2	6	0.158

Table S16: Independent admission under warm anomalies, compound low-flow/heat overlap and the independently published event catalog. Flow missing on required gap-event days is explicitly unknown, not absence of compound stress. The project event threshold requires three days; published episodes require two. These are outcome strata, never deployment features.

Condition	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
False	Retro	foen	F0	229	12	7	0.115
True	Retro	foen	F0	146	21	7	0.243
flow_incomplete_on_gap_event_days	Retro	foen	F0	13	3	7	0.400
gap_intersects_compound_event	Retro	foen	F0	58	17	7	0.489
no_gap_thermal_event_overlap	Retro	foen	F0	262	10	7	0.095
observed_no_compound_event_overlap	Retro	foen	F0	42	3	7	0.123
False	Retro	foen	F0	375	33	7	0.150
False	Retro	foen	F1	191	6	7	0.078
True	Retro	foen	F1	127	7	7	0.113
flow_incomplete_on_gap_event_days	Retro	foen	F1	11	1	7	0.231
gap_intersects_compound_event	Retro	foen	F1	42	7	7	0.381
no_gap_thermal_event_overlap	Retro	foen	F1	227	4	7	0.047
observed_no_compound_event_overlap	Retro	foen	F1	38	1	7	0.059
False	Retro	foen	F1	318	13	7	0.087
False	Retro	hubeau	F0	388	14	12	0.075
True	Retro	hubeau	F0	96	16	7	0.284

Condition	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
flow_incomplete_on_gap_event_days	Retro	hubeau	F0	141	14	12	0.203
no_gap_thermal_event_overlap	Retro	hubeau	F0	343	16	12	0.091
False	Retro	hubeau	F0	484	30	12	0.115
False	Retro	hubeau	F1	286	3	12	0.031
True	Retro	hubeau	F1	76	5	7	0.123
flow_incomplete_on_gap_event_days	Retro	hubeau	F1	94	3	12	0.083
no_gap_thermal_event_overlap	Retro	hubeau	F1	268	5	12	0.049
False	Retro	hubeau	F1	362	8	12	0.049
False	Prospective	foen	F0	36	0	7	0.000
True	Prospective	foen	F0	46	8	7	0.304
flow_incomplete_on_gap_event_days	Prospective	foen	F0	10	2	7	0.333
gap_intersects_compound_event	Prospective	foen	F0	13	4	7	0.556
no_gap_thermal_event_overlap	Prospective	foen	F0	54	2	7	0.120
observed_no_compound_event_overlap	Prospective	foen	F0	5	0	6	0.000
False	Prospective	foen	F0	82	8	7	0.186
False	Prospective	foen	F1	32	0	7	0.000
True	Prospective	foen	F1	32	1	7	0.094
flow_incomplete_on_gap_event_days	Prospective	foen	F1	7	0	7	0.000
gap_intersects_compound_event	Prospective	foen	F1	7	1	7	0.273
no_gap_thermal_event_overlap	Prospective	foen	F1	46	0	7	0.000

Condition	Time	Provider	Mode	Admit	Fail	Groups	Risk upper
observed_no_compound_event_overlap	Prospective	foen	F1	4	0	6	0.000
False	Prospective	foen	F1	64	1	7	0.045
False	Prospective	hubeau	F0	18	1	4	0.176
True	Prospective	hubeau	F0	24	2	4	0.273
flow_incomplete_on_gap_event_days	Prospective	hubeau	F0	16	3	6	0.409
no_gap_thermal_event_overlap	Prospective	hubeau	F0	26	0	6	0.000
False	Prospective	hubeau	F0	42	3	6	0.167
False	Prospective	hubeau	F1	14	0	4	0.000
True	Prospective	hubeau	F1	22	2	4	0.300
flow_incomplete_on_gap_event_days	Prospective	hubeau	F1	12	2	6	0.364
no_gap_thermal_event_overlap	Prospective	hubeau	F1	24	0	6	0.000
False	Prospective	hubeau	F1	36	2	6	0.158

Table S17: Every independent new-station component for the uncertain-window policy. Counts expose concentrated failures that pooled risk can hide. Repeated masks and years within one river are not independent Bernoulli trials: a zero numerator is not a safety certificate and no case-binomial confidence interval is claimed.

Station	Time	Mode	Pool	Admit	Fail	Risk
CH_2347	Retro	F0	94	58	5	0.086
CH_2609	Retro	F0	94	45	2	0.044
CH_2179	Retro	F0	97	56	2	0.036
CH_2612	Retro	F0	97	47	3	0.064
CH_2604	Retro	F0	94	50	5	0.100

Station	Time	Mode	Pool	Admit	Fail	Risk
CH_2608	Retro	F0	97	51	0	0.000
CH_2327	Retro	F0	96	68	16	0.235
CH_2347	Retro	F1	94	49	1	0.020
CH_2609	Retro	F1	94	40	2	0.050
CH_2179	Retro	F1	97	48	0	0.000
CH_2612	Retro	F1	97	40	0	0.000
CH_2604	Retro	F1	94	39	1	0.026
CH_2608	Retro	F1	97	42	0	0.000
CH_2327	Retro	F1	96	60	9	0.150
MC_5099170	Retro	F0	67	52	1	0.019
PY_5180550	Retro	F0	58	46	4	0.087
MC_5096050	Retro	F0	59	46	0	0.000
PY_5180295	Retro	F0	88	60	4	0.067
AL_6078200	Retro	F0	55	32	6	0.188
PY_5218700	Retro	F0	24	24	0	0.000
AL_6152400	Retro	F0	57	29	9	0.310
PY_5200050	Retro	F0	44	36	0	0.000
PY_6175400	Retro	F0	51	29	5	0.172
PY_5172350	Retro	F0	71	49	1	0.020
AL_6143650	Retro	F0	69	60	0	0.000
PY_5110000	Retro	F0	57	21	0	0.000
MC_5099170	Retro	F1	67	42	0	0.000
PY_5180550	Retro	F1	58	37	1	0.027
MC_5096050	Retro	F1	59	37	0	0.000
PY_5180295	Retro	F1	88	26	0	0.000
AL_6078200	Retro	F1	55	25	2	0.080
PY_5218700	Retro	F1	24	20	0	0.000
AL_6152400	Retro	F1	57	19	2	0.105
PY_5200050	Retro	F1	44	24	0	0.000
PY_6175400	Retro	F1	51	23	3	0.130
PY_5172350	Retro	F1	71	39	0	0.000
AL_6143650	Retro	F1	69	51	0	0.000

Station	Time	Mode	Pool	Admit	Fail	Risk
PY_5110000	Retro	F1	57	19	0	0.000
CH_2347	Prospective	F0	26	11	0	0.000
CH_2609	Prospective	F0	24	8	0	0.000
CH_2179	Prospective	F0	25	11	1	0.091
CH_2612	Prospective	F0	25	11	4	0.364
CH_2604	Prospective	F0	24	13	2	0.154
CH_2608	Prospective	F0	25	9	0	0.000
CH_2327	Prospective	F0	24	19	1	0.053
CH_2347	Prospective	F1	26	8	0	0.000
CH_2609	Prospective	F1	24	6	0	0.000
CH_2179	Prospective	F1	25	8	0	0.000
CH_2612	Prospective	F1	25	6	0	0.000
CH_2604	Prospective	F1	24	11	1	0.091
CH_2608	Prospective	F1	25	7	0	0.000
CH_2327	Prospective	F1	24	18	0	0.000
MC_5099170	Prospective	F0	11	7	0	0.000
PY_5180550	Prospective	F0	6	5	0	0.000
PY_5180295	Prospective	F0	12	4	1	0.250
AL_6078200	Prospective	F0	26	13	0	0.000
AL_6152400	Prospective	F0	24	10	2	0.200
PY_5172350	Prospective	F0	6	3	0	0.000
MC_5099170	Prospective	F1	11	5	0	0.000
PY_5180550	Prospective	F1	6	5	0	0.000
PY_5180295	Prospective	F1	12	3	0	0.000
AL_6078200	Prospective	F1	26	12	0	0.000
AL_6152400	Prospective	F1	24	9	2	0.222
PY_5172350	Prospective	F1	6	2	0	0.000

Table S18: Constructed population stress test, not observed river failure rates. Groups are independent with perfectly dependent failures within groups, true failure probability 0.06 and uninformative scores. Empty admissions are not zero risk. The rigorous bounded-group test addresses a group-expectation ratio under its assumptions, not every catchment or arbitrary domain shift.

scenario	n repetitions	empirical violation probability	rigorous violation probability	empirical admission probability	rigorous admission probability
forty_perfectly_correlated	1000	0.085	0.000	0.085	0.000
hundred_perfectly_correlated	1000	0.047	0.002	0.047	0.002
rare_bad_group	1000	0.000	0.000	0.450	0.000
twelve_perfectly_correlated	1000	0.454	0.000	0.454	0.000

Table S19: Opened-source component-held-out event/cofailure assay. Exactly the admission-trained heads and thresholds are applied without adaptation. Values average the three dependent split sensitivities. Before/during/after masks are relative to the same longest training-threshold episode. The new European component bundles are single-station and cannot by themselves identify donor-cofailure sensitivity; this nondegenerate multi-station test supplies that diagnostic, not independent confirmation.

Gap phase	Inputs	Removed share	Retention	Selected risk	Candidates
after	Retro	0.000	0.858	0.025	7938
before	Retro	0.000	0.864	0.025	6932
during	Retro	0.000	0.766	0.062	7017
after	Retro	0.500	0.837	0.024	7938
before	Retro	0.500	0.839	0.027	6932

Gap phase	Inputs	Removed share	Retention	Selected risk	Candidates
during	Retro	0.500	0.739	0.072	7017
after	Retro	1.000	0.764	0.038	7938
before	Retro	1.000	0.780	0.034	6932
during	Retro	1.000	0.663	0.112	7017
after	Past-local	0.000	0.846	0.019	7323
before	Past-local	0.000	0.858	0.013	6141
during	Past-local	0.000	0.763	0.051	6482
after	Past-local	0.500	0.824	0.018	7323
before	Past-local	0.500	0.827	0.017	6141
during	Past-local	0.500	0.724	0.055	6482
after	Past-local	1.000	0.752	0.025	7323
before	Past-local	1.000	0.774	0.023	6141
during	Past-local	1.000	0.654	0.092	6482

Table S20: No-screen composition rule on independent natural-calendar replay. Only French records contribute qualifying calendars here; there is no independent Swiss documented-mean calendar estimate. Bin boundaries were inherited from opened analyses; cell risks were fitted on opened-source empirical geometries and fixed before new-test replay. Advice based on a point forecast is not certification; source-only prediction intervals omit unmodelled domain shift and one target component cannot establish safety.

Component	Time	Masks	Station-years	Predicted risk	Observed risk	No-screen advice
external_05987568be9b240d	Retro	30	6	0.286	0.133	False
external_23920287a08e1d16	Retro	18	9	0.079	0.000	False
external_532dbe5907561a81	Retro	24	8	0.415	0.333	False

Component	Time	Masks	Station-years	Predicted risk	Observed risk	No-screen advice
external_7d32d6204897376e	Retro	8	4	0.276	0.000	False
external_9c4bbb8c94b72fa2	Retro	20	5	0.415	0.700	False
external_e2ef71f685704435	Retro	8	4	0.415	0.125	False
external_e3d145690e8a064e	Retro	16	8	0.218	0.500	False
external_ed65c2242ac535c3	Retro	22	11	0.015	0.000	True
external_f80499d88ac9bced	Retro	12	6	0.415	0.000	False
external_f9dc2e50bdc51aa2	Retro	10	5	0.215	0.100	False
external_05987568be9b240d	Prospective	3	1	0.257	0.000	False
external_23920287a08e1d16	Prospective	2	1	0.149	0.000	False
external_532dbe5907561a81	Prospective	3	1	0.475	0.667	False
external_9c4bbb8c94b72fa2	Prospective	8	2	0.475	0.500	False
external_ed65c2242ac535c3	Prospective	2	1	0.040	0.000	True

Table S21: Reconstruction references on identical replicate-zero crossed cases. Availability is explicit, and the fixed policy MAE is recomputed on each method's identical supported subset. Inputs differ, so differences are not attributed solely to architecture. CSDI is an old-source-trained, fixed eight-draw retrospective univariate reference, not a claim to optimized state of the art. Air2stream parameters and discharge normalization exclude the target year and are strictly prior prospectively; every gap starts from an observed left anchor. No unsupported prediction is counted as a success.

Reconstructor	Time	Information	Available	H30 MAE C	Matched policy C	Failure share
CSDI: median series	Retro	Target only	1701	0.453	0.311	0.304
CSDI: median H30	Retro	Target only	1701	0.434	0.311	0.290
Daily climate	Retro	Target only	1701	0.401	0.311	0.297
Smooth climate	Retro	Target only	1701	0.399	0.311	0.305
Interpolation	Retro	Target only	1701	0.428	0.311	0.273
Neighbour ridge	Retro	Target + neighbours	0	–	–	–
Air ridge	Retro	Target + air	1544	0.280	0.280	0.215
air2stream3	Retro	Target + air	1544	0.254	0.280	0.192
air2stream8	Retro	Target + air + flow	821	0.203	0.342	0.157
air2stream3	Prospective	Target + air	185	0.277	0.332	0.211
air2stream8	Prospective	Target + air + flow	135	0.173	0.368	0.104

Table S22: Raw 90 percent conditional-trajectory band diagnostics. These are uncalibrated scenario bands for the fixed priority point, not valid confidence intervals or selected-risk certificates. Missing bands remain missing. The conditional residual bootstrap does not capture all bias or transfer uncertainty.

Assay	Time	Provider	Candidates	Bands	Coverage	Width C
admission	Retro	foen	669	669	0.716	0.798
admission	Retro	hubeau	700	700	0.813	0.546
crossed	Retro	foen	6718	6718	0.730	0.770
crossed	Retro	hubeau	5631	5631	0.831	0.535
events	Retro	foen	582	582	0.753	0.374
events	Retro	hubeau	492	492	0.904	0.326
real_calendars	Retro	hubeau	168	168	0.804	0.738
admission	Prospective	foen	173	173	0.607	0.943
admission	Prospective	hubeau	85	85	0.682	0.780
crossed	Prospective	foen	1790	1790	0.684	0.895
crossed	Prospective	hubeau	655	655	0.794	0.710
events	Prospective	foen	182	182	0.780	0.349
events	Prospective	hubeau	67	67	0.806	0.385
real_calendars	Prospective	hubeau	18	18	0.667	1.001

Table S23: Full original-candidate locally anomalous-hot decisions, fixed source uncertain-window policy. False-hot cost is one; costs were declared before new test. Negative differences favor screening. Always-unknown cost equals the declared abstention cost, with no magnitude estimate or finite selected risk. This stronger deterministic comparator was added in the post-test scientific audit without changing any decision. Always-hot/cool costs and the original descriptive component-bootstrap intervals remain in the source CSV; no simultaneous guarantee or measured agency preference is claimed.

Time	Provider	Miss cost	Unknown cost	Screen loss	vs all recon.	vs all unknown
Retro	foen	1.000	0.000	0.019	-0.072	0.019
Retro	foen	1.000	0.010	0.024	-0.067	0.014
Retro	foen	1.000	0.025	0.030	-0.061	0.005
Retro	foen	1.000	0.050	0.041	-0.050	-0.009
Retro	foen	1.000	0.100	0.063	-0.028	-0.037
Retro	foen	1.000	0.250	0.129	0.038	-0.121
Retro	foen	5.000	0.000	0.097	-0.359	0.097
Retro	foen	5.000	0.010	0.102	-0.354	0.092
Retro	foen	5.000	0.025	0.108	-0.348	0.083
Retro	foen	5.000	0.050	0.119	-0.337	0.069
Retro	foen	5.000	0.100	0.141	-0.315	0.041
Retro	foen	5.000	0.250	0.207	-0.249	-0.043
Retro	foen	10.000	0.000	0.194	-0.717	0.194
Retro	foen	10.000	0.010	0.199	-0.713	0.189
Retro	foen	10.000	0.025	0.205	-0.707	0.180
Retro	foen	10.000	0.050	0.216	-0.696	0.166
Retro	foen	10.000	0.100	0.238	-0.674	0.138
Retro	foen	10.000	0.250	0.304	-0.608	0.054
Retro	hubeau	1.000	0.000	0.060	-0.079	0.060
Retro	hubeau	1.000	0.010	0.063	-0.075	0.053
Retro	hubeau	1.000	0.025	0.068	-0.071	0.043

Time	Provider	Miss cost	Unknown cost	Screen loss	vs all recon.	vs all unknown
Retro	hubeau	1.000	0.050	0.075	-0.063	0.025
Retro	hubeau	1.000	0.100	0.091	-0.048	-0.009
Retro	hubeau	1.000	0.250	0.137	-0.001	-0.113
Retro	hubeau	5.000	0.000	0.203	-0.210	0.203
Retro	hubeau	5.000	0.010	0.206	-0.207	0.196
Retro	hubeau	5.000	0.025	0.211	-0.202	0.186
Retro	hubeau	5.000	0.050	0.218	-0.195	0.168
Retro	hubeau	5.000	0.100	0.234	-0.179	0.134
Retro	hubeau	5.000	0.250	0.280	-0.133	0.030
Retro	hubeau	10.000	0.000	0.381	-0.374	0.381
Retro	hubeau	10.000	0.010	0.385	-0.371	0.375
Retro	hubeau	10.000	0.025	0.389	-0.367	0.364
Retro	hubeau	10.000	0.050	0.397	-0.359	0.347
Retro	hubeau	10.000	0.100	0.412	-0.343	0.312
Retro	hubeau	10.000	0.250	0.459	-0.297	0.209
Prospective	foen	1.000	0.000	0.012	-0.098	0.012
Prospective	foen	1.000	0.010	0.017	-0.093	0.007
Prospective	foen	1.000	0.025	0.025	-0.085	-0.000
Prospective	foen	1.000	0.050	0.038	-0.072	-0.012
Prospective	foen	1.000	0.100	0.064	-0.046	-0.036
Prospective	foen	1.000	0.250	0.143	0.033	-0.107
Prospective	foen	5.000	0.000	0.058	-0.491	0.058
Prospective	foen	5.000	0.010	0.063	-0.486	0.053
Prospective	foen	5.000	0.025	0.071	-0.478	0.046
Prospective	foen	5.000	0.050	0.084	-0.465	0.034
Prospective	foen	5.000	0.100	0.110	-0.439	0.010
Prospective	foen	5.000	0.250	0.189	-0.360	-0.061
Prospective	foen	10.000	0.000	0.116	-0.983	0.116
Prospective	foen	10.000	0.010	0.121	-0.977	0.111
Prospective	foen	10.000	0.025	0.129	-0.970	0.104
Prospective	foen	10.000	0.050	0.142	-0.956	0.092

Time	Provider	Miss cost	Unknown cost	Screen loss	vs all recon.	vs all unknown
Prospective	foen	10.000	0.100	0.168	-0.930	0.068
Prospective	foen	10.000	0.250	0.247	-0.851	-0.003
Prospective	hubeau	1.000	0.000	0.024	-0.188	0.024
Prospective	hubeau	1.000	0.010	0.029	-0.183	0.019
Prospective	hubeau	1.000	0.025	0.036	-0.176	0.011
Prospective	hubeau	1.000	0.050	0.049	-0.163	-0.001
Prospective	hubeau	1.000	0.100	0.074	-0.138	-0.026
Prospective	hubeau	1.000	0.250	0.150	-0.062	-0.100
Prospective	hubeau	5.000	0.000	0.024	-0.565	0.024
Prospective	hubeau	5.000	0.010	0.029	-0.560	0.019
Prospective	hubeau	5.000	0.025	0.036	-0.552	0.011
Prospective	hubeau	5.000	0.050	0.049	-0.539	-0.001
Prospective	hubeau	5.000	0.100	0.074	-0.514	-0.026
Prospective	hubeau	5.000	0.250	0.150	-0.438	-0.100
Prospective	hubeau	10.000	0.000	0.024	-1.035	0.024
Prospective	hubeau	10.000	0.010	0.029	-1.030	0.019
Prospective	hubeau	10.000	0.025	0.036	-1.023	0.011
Prospective	hubeau	10.000	0.050	0.049	-1.010	-0.001
Prospective	hubeau	10.000	0.100	0.074	-0.985	-0.026
Prospective	hubeau	10.000	0.250	0.150	-0.909	-0.100

Table S24: Locally hot and cool refusals, selected-sample frequency error and original-pool unknown-label bounds. Bounds allow every unknown label to be cool or hot and are not statistical confidence intervals; classification mistakes among judged cases can still affect them. Magnitude and hot-label admission are distinct outputs. Declaring an observed hot lower bound never certifies a cool record.

Time	Provider	Decision	Hot refusal	Cool refusal	Freq. error	Pool lower	Pool upper
Retro	foen	Magnitude	0.464	0.434	-0.042	0.070	0.510
Retro	foen	Hot-label rescue	0.446	0.434	-0.037	0.073	0.510
Retro	hubeau	Magnitude	0.355	0.296	-0.031	0.129	0.437
Retro	hubeau	Hot-label rescue	0.349	0.296	-0.030	0.130	0.437
Prospective	foen	Magnitude	0.488	0.538	-0.005	0.116	0.642
Prospective	foen	Hot-label rescue	0.465	0.538	0.004	0.121	0.642
Prospective	hubeau	Magnitude	0.400	0.580	0.136	0.271	0.776
Prospective	hubeau	Hot-label rescue	0.371	0.580	0.146	0.282	0.776

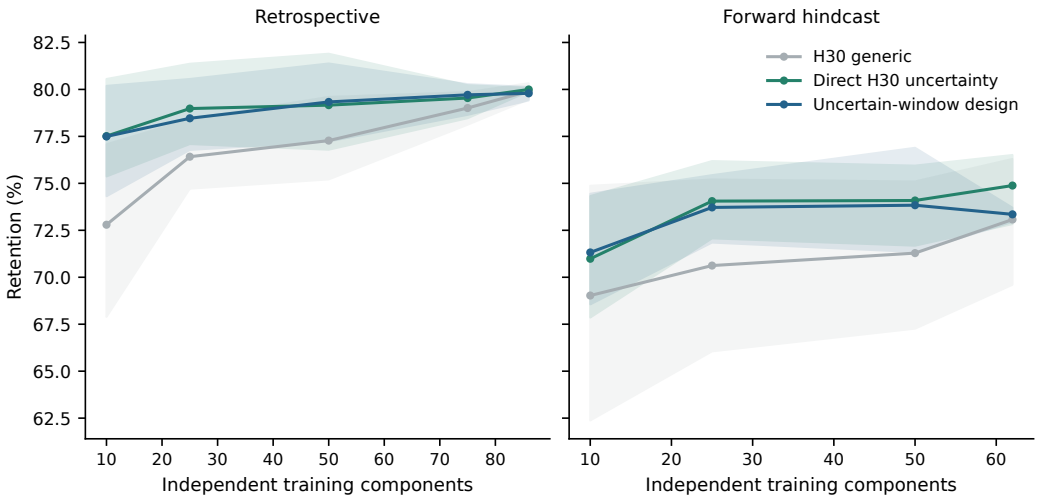


Figure S1. Repeated learning curves. Shading spans the tenth to ninetieth percentiles over training subsets, not new independent rivers or confidence limits. Calibration/evaluation components are fixed. Prospective global fitting/calibration use only years through 2018.

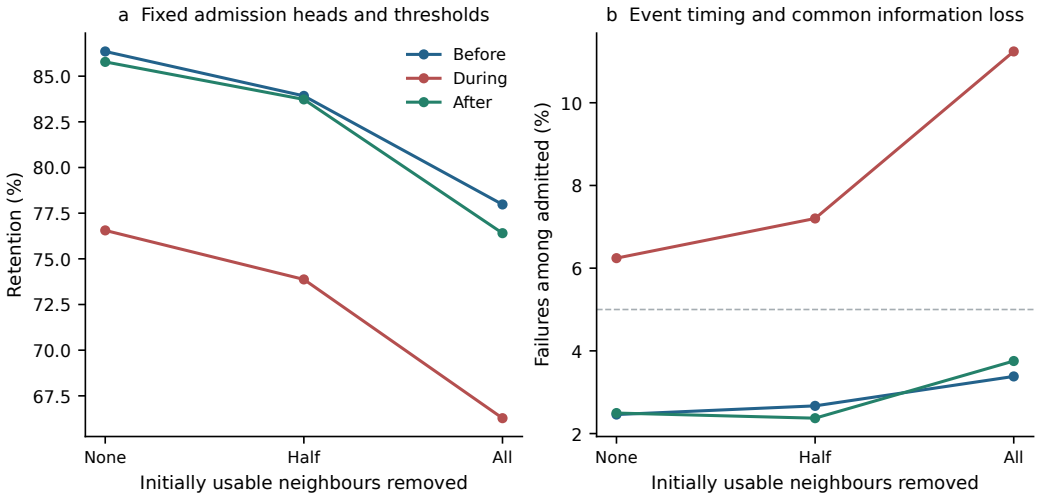


Figure S2. Nondegenerate opened-source event/cofailure stress with unchanged admission heads and thresholds. Values average three split sensitivities, not independent experiments. Timing is relative to the same thermal episode. The independent European station bundles lack donor variation, so this plot is not independent target confirmation.